LECTURE 1 · Introduction & probability

Big picture

- Info theory = art of redundancy: **remove** it when it hurts (compression/source coding), **add** it when it helps (channel coding). Info & statistics = two sides of one coin.
- Source coding desiderata: code is **short** & **easily decodable**. Frequent symbols → shorter words (e.g. 1.56 bit/pixel vs 3 bit one-hot).
- **BSC**: P(y=0|x=0)=P(y=1|x=1)=1−f; flips w.p. f. **Repetition code R3**: send each bit 3×; optimal decoder = Bayes $\hat{s}=\text{argmax}_s$ P(s|r)P(s)… = **majority vote**. Bit error $p_b=3f^2(1-f)+f^3 \approx 3f^2$: error O(f)→O(f²) but rate ÷3. **Shannon**: error-free comms at non-zero rate is possible (rates below capacity C).

Maths toolkit

- **Logs**: $x=b^{\log_b x}$; log(xy)=log x+log y; log x^n=n log x; $\log_b x=\log_a x / \log_a b$; log 1=0; log(x+y) does **not** simplify. log≡$\log_2$, ln≡$\log_e$, ln2≈0.69, log(1/8)=−3.
- **Convex** (∪): $f(\lambda x_1+(1-\lambda)x_2) \le \lambda f(x_1)+(1-\lambda)f(x_2)$. Prove via f″≥0 or convexity-preserving ops (affine map f(Ax+b); non-neg. weighted sum $\Sigma w_i f_i$, $w_i \ge 0$). log x and −x log x are concave (∩).
- **Jensen**: f convex ⇒ E[f(X)] ≥ f(E[X]) (reversed for concave, e.g. E[log X] ≤ log E[X]).
- **Random variable** X: Ω→E measurable; sample space Ω, event space E, P(ω)≥0, P(Ω)=1. Discrete: PMF $\Sigma_x P(x)=1$. Continuous: PDF $\int p(x)dx=1$ (a density **can exceed 1**; $\int_a^b p \le 1$). Notation: X rv, x realisation, X alphabet, X~p.
- $E_p[f(X)]=\Sigma_x p(x)f(x)$; $\text{Var}(X)=E[(X-\mu)^2]=E[X^2]-\mu^2$. **i.i.d.** = independent + identically distributed.
- **Normal** $N(\mu,\Sigma)$: $p(x)=(2\pi)^{-d/2}|\Sigma|^{-1/2}\exp(-\tfrac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu))$; within ±1σ/2σ/3σ: 68.3% / 95.5% / 99.7%.
- Function f is **injective** if $f(x_1)=f(x_2)\Rightarrow x_1=x_2$; **bijective** = injective + surjective ⇒ invertible. Limits: ε–δ definition; $\lim_{x\to 0^+} \ln x=-\infty$.
- **Sets**: ∈ membership, ⊂ subset, S^c complement, |S| cardinality, ∪ union, ∩ intersection. **Course map**: Part I source coding (compression) · Part II channel coding (transmission) · Part III statistics & advanced (inference, estimation, information decomposition).

LECTURE 2 · Entropy & source coding

Definitions

- **Surprisal** h(x)=log(1/p(x))=−log p(x). **Entropy** $H(X)=\Sigma_x p(x)\log(1/p(x))=E[h(X)]$. Units: $\log_2$→bits, ln→nats.
- **Binary entropy** $H_2(p)=-p\log p-(1-p)\log(1-p)$; X~Bernoulli(p); $H_2(0.5)=1$, $H_2(0.1)=0.47$, $H_2(0.2)=0.72$.
- **Bounds** (discrete): $0 \le H(X) \le \log|X|$. Min at Kronecker delta; max at uniform. Convention 0 log 0=0.
- **Independent**: h(x,y)=h(x)+h(y); H(X,Y)=H(X)+H(Y) (additive).
- **Shannon axioms** for surprisal i(p): (1) i(1)=0; (2) di/dp ≤ 0; (3) i(pq)=i(p)+i(q). Only solution: $i(p)=-\log_b p$.
- Uniform: h(x)=# yes/no questions (game 63 → 6 bit; submarine: total info always log 64=6 bit however long it takes). **Design experiments with evenly probable outcomes** to maximise information.

Continuous / Gaussian

- Discretise bins Δ: $H(X^\Delta)=-\Sigma\Delta f(x_i)\log f(x_i)-\log\Delta \to$ diverges. **Differential entropy** $H(X)=-\int f(x)\log f(x)dx$ (drops log Δ): can be **negative**; U[0,a]: H=log a; 2^H=volume.
- Gaussian surprisal: $h(x)=\ln\sigma+\ln\sqrt{(2\pi)}+(x-\mu)^2/(2\sigma^2)$ (scaled MSE). **$H(N(\mu,\Sigma))=\tfrac{1}{2}\ln|2\pi e\Sigma|$** (1-D: $\tfrac{1}{2}\ln(2\pi e\sigma^2)$); entropy = generalised variance.

Typical sets & the source coding theorem

- **wLLN**: $Y_n=(1/n)\Sigma X_i$: E=μ, $\text{Var}=\sigma^2/n$; $Y_n \to \mu$ in probability. High-dim Gaussian: points concentrate in a shell, $\text{Std}(r^2)/E[r^2]=O(1/\sqrt{n})$.
- **AEP**: X_i i.i.d.~p ⇒ $-(1/n)\log p(x_1\dots x_n) \to H(X)$.
- **Typical set** $A_\varepsilon^{(n)}$: $2^{-n(H+\varepsilon)} \le p(x^n) \le 2^{-n(H-\varepsilon)}$. Four properties: **(1)** $x^n\in A \Rightarrow -(1/n)\log p(x^n)\in[H-\varepsilon,H+\varepsilon]$ (all ≈ equiprobable, $p\approx 2^{-nH}$); **(2)** $P(x^n\in A)>1-\varepsilon$; **(3)** $|A| \le 2^{n(H+\varepsilon)}$; **(4)** $|A| \ge (1-\varepsilon)2^{n(H-\varepsilon)}$. ($2^n$ strings but only $\approx 2^{nH}$ typical.) Biased coin p: typical strings have ≈ fraction p of ones.
- **Source Coding Theorem (SCT)**: X^n i.i.d.~p. ∀ε>0, large n: ∃ invertible code with $L/n=E[\ell(X^n)]/n \le H(X)+\varepsilon$. **Converse**: no invertible code with L < H(X). (Achievability + converse.)
- *Achievability proof*: flag bit 0 + index in A ($\ell \le n(H+\varepsilon)+2$) if $x^n\in A$; flag 1 + index in X^n ($\ell \le n\log|X|+2$) otherwise ⇒ $L \le n(H+\varepsilon')$. *Converse*: a subset of size $2^{n(H-\kappa)}$ has prob $\approx 2^{-n\kappa}\to 0$.
- **Code** C:X→D*, ℓ(x) length, L=E[ℓ]; binary if |D|=2. Raw bit content log|X|; length $\lceil\log|X|\rceil$ (the "extra bit") if |X| not a power of 2. Entropy as volume: H∝log V+C (Gaussian).
- **Surprisal table** h(p)=log(1/p): p=0.5→1.0 bit; 0.2→2.3; 0.1→3.3; 0.01→6.6; 0.001→10.0 (unlikely outcomes carry more information). $H_2(0.001)=0.011$, $H_2(0.01)=0.081$. Game 63: Q's "x≥32?", "x mod 32≥16?",… ⇒ 6-bit binary code (35⇒100011). Submarine: miss 1st shot gives log(64/63)=0.0227 bit; telescoping Σ=log 64=6 bit.

LECTURE 3 · Symbol codes, KL & ML

Types of code (nested)

- **No free lunch**: no compressor shortens all strings and is invertible ⇒ **lossy** (maps some strings together) or **lossless** (lengthens some).
- **Nonsingular**: $x\neq x' \Rightarrow C(x)\neq C(x')$. **Extension** $C^*(x_1\dots x_n)=C(x_1)\dots C(x_n)$. **Uniquely decodable (UD)**: C* nonsingular. **Prefix** (instantaneous / self-punctuating): no codeword is the start of another. Prefix ⊂ UD ⊂ nonsingular. Morse = nonsingular, not UD (needs punctuation). Invertible ≠ easy to invert (RSA).
- Prefix codes = trees (leaves = codewords). Complete D-ary tree has $D^{\ell\max}$ leaves; adding a word removes its descendants.
- **Kraft inequality**: lengths of any D-ary prefix code satisfy $\mathbf{\Sigma_i D^{-\ell i} \le 1}$; conversely any lengths satisfying it ⇒ ∃ prefix code. Also holds for UD codes. ("codeword supermarket": short words cost more, fixed budget.) *Proof*: ℓ_i word has $D^{\ell\max-\ell i}$ descendants at level ℓmax, total ≤ $D^{\ell\max}$.

Optimal lengths & Huffman

- **Lagrange**: min f(x) s.t. g(x)=0 → L=f+λ·g, solve $\nabla_{x,\lambda} L=0$ (k+c equations).
- Min $\Sigma p_i\ell_i$ s.t. $\Sigma D^{-\ell i}=1 \Rightarrow \mathbf{\ell_i^*=-\log_D p_i}$, $\mathbf{L^*=H_D(X)}$: entropy = min average length of any symbol code (2nd derivative ≥0 ⇒ global min). Duality: PDF $p_i \leftrightarrow$ lengths ℓ_i; implicit probs $p_i \propto D^{-\ell i}$. Shannon code (mock exam): $\ell_i=\lceil -\log p_i\rceil$; check $\Sigma 2^{-\ell i}\le 1$ (Kraft), $E[\ell]=\Sigma p_i\ell_i \ge H$.
- **Huffman**: repeat: take two least-probable symbols, merge into one (tree); read codewords from tree. **Optimal**: $L(C^*) \le L(C')$ for any UD code C′. E.g. p={.25,.25,.2,.15,.15}: a→00, b→10, e→011.

Cross-entropy & KL divergence

- **Cross-entropy** $H(p,q)=-\Sigma p(x)\log q(x)$ = avg length of a q-optimal code used on p-data.
- $\mathbf{D_{KL}(p\|q)=\Sigma p(x)\log(p(x)/q(x)) = H(p,q)-H(p)}$ = extra cost of using the wrong code. **Gibbs' inequality**: $D_{KL} \ge 0$, =0 iff p=q ∀x (proof: $-D_{KL}=\Sigma p\log(q/p) \le \log\Sigma q = 0$ by Jensen, log concave).
- Properties: **asymmetric** ($D_{KL}(p\|q)\neq D_{KL}(q\|p)$); =∞ if ∃x: q(x)=0,p(x)>0; convex in q; $D_{KL}(p\|u)=\log|X|-H(p)$ ⇒ H≤log|X|. Continuous: log Δ terms cancel ⇒ $D_{KL}=\int f\log(f/g)\ge 0$. Conditional KL: $D_{KL}(p(X|Y)\|q(X|Y))=\Sigma_y p(y)D_{KL}(p(X|y)\|q(X|y)) \ge 0$.
- **Gaussian KL**: $D_{KL}(p\|q)=\tfrac{1}{2}[\sigma_p^2/\sigma_q^2-1-\ln(\sigma_p^2/\sigma_q^2)+(\mu_p-\mu_q)^2/\sigma_q^2]$. $H(p,q)=\tfrac{1}{2}\ln(2\pi\sigma_q^2)+[\sigma_p^2+(\mu_p-\mu_q)^2]/(2\sigma_q^2)$. Narrow Gaussians: KL grows faster with Δμ.

Maximum likelihood & compression=prediction

- **MLE** $\theta^*=\text{argmax}_\theta p(x;\theta)$; i.i.d.: $L(\theta)=(1/n)\Sigma\ln p(x_i;\theta) \to E_p[\ln p(x;\theta)]$ (n→∞). Gaussian mean: $\mu^*=(1/n)\Sigma x_i$. **ML ⇔ minimum code length** (min KL).
- Code $\ell(x) \Rightarrow$ implicit $q(x)=2^{-\ell(x)}/z$, $z=\Sigma 2^{-\ell}$ (Kraft number, z≤1). Extra cost $E_p[\ell-\ell^*]=\mathbf{\log(1/z)+D_{KL}(p\|q)} \ge 0$; zero iff Kraft equality & p=q. "A good compressor is a good predictor."
- **SCT for symbol codes**: for any p(x) ∃ code with L close to H(X) (achievability) and no code with L<H(X) (converse). Block codes (AEP) need infinite n and enumerating 2^{nH} typical strings ⇒ impractical; symbol codes are practical.
- **Worked wrong-code example**: believed q={½,¼,⅛,⅛}, code {0,10,110,111} (expected 1.75 bit); true p={0,⅓,⅓,⅓} ⇒ actual avg length −Σp log q=⅓·2+⅓·3+⅓·3 ≈ 2.66 bit (cross-entropy H(p,q)). Gaussian-mean ML: $dL/d\mu=(1/n\sigma^2)\Sigma(x_i-\mu)=0 \Rightarrow \mu^*=\bar{x}$.

LECTURE 4 · Joint entropy & MI

Multivariate probability

- Marginal $p(x)=\Sigma_y p(x,y)$; conditional p(x|y)=p(x,y)/p(y) (undefined if p(y)=0). **Product** p(x,y)=p(x|y)p(y); **sum** $p(x)=\Sigma_y p(x|y)p(y)$; **Bayes** p(x|y)=p(y|x)p(x)/p(y).
- **Independence**: marginal $X\perp\!\!\!\perp Y \Leftrightarrow p(x,y)=p(x)p(y)$ (⇒p(X|Y)=p(X)); conditional $X\perp\!\!\!\perp Y|Z \Leftrightarrow p(x,y|z)=p(x|z)p(y|z)$. Conditioning can create/destroy dependence: Y=f(X), Z=g(Y): $\neg(X\perp\!\!\!\perp Z)$ but $X\perp\!\!\!\perp Z|Y$; X,Y~B(½), Z=X+Y mod 2: $X\perp\!\!\!\perp Z$ but $\neg(X\perp\!\!\!\perp Z|Y)$.
- **Bayesian network**: $p(X)=\Pi_k p(X_k|pa_k)$ (e.g. p(X)p(Y|X)p(Z|X)p(U|Y,Z)). **Markov chain** X−Y−Z: $p(X_t|X_{t-1}\dots X_1)=p(X_t|X_{t-1})$ (order p: depends on last p).

Entropies

- **Joint** H(X,Y)=−Σp(x,y)log p(x,y) (symmetric). **Conditional** $H(X|Y)=\Sigma_y p(y)H(X|Y=y)$, $H(X|Y=y)=-\Sigma_x p(x|y)\log p(x|y)$. Props: $0\le H(X|Y)\le\log|X|$; =0 iff X=f(Y); =H(X) iff $X\perp\!\!\!\perp Y$.
- **Entropy chain rule**: **H(X,Y)=H(X|Y)+H(Y)=H(Y|X)+H(X)**. **Isomorphism** (bijection X=f(Y)): H(X)=H(Y) (discrete). Example: H(X)=1, H(Y)=1.41, H(X|Y)=0.84, H(Y|X)=1.25, H(X,Y)=2.25 bit.
- **Stationary process**: $p(X_1\dots X_t)=p(X_{1+s}\dots X_{t+s})$. **Entropy rate** h=lim (1/n)$H(X_1\dots X_n)$ = lim $H(X_n|X_1\dots X_{n-1})$ (innovation) – equal for stationary. More conditioning ⇒ lower entropy; curve plateaus at h (finite memory = Markov order). Coin in a box (flip prob p): $h=H_2(p)$. Stream codes (Lempel–Ziv) ↔ symbol codes.

Mutual information

- **$\mathbf{I(X;Y)=\Sigma p(x,y)\log[p(x,y)/(p(x)p(y))] = H(X)+H(Y)-H(X,Y) = H(X)-H(X|Y) = H(Y)-H(Y|X) = D_{KL}(p(X,Y)\|p(X)p(Y))}$**. Symmetric; ≥0; =0 iff $X\perp\!\!\!\perp Y$; ≤min{H(X),H(Y)} (discrete). Avg reduction in uncertainty.
- **Information doesn't hurt**: H(X|Y) ≤ H(X) (but H(X|Y=y) can exceed H(X)).
- **CMI** $I(X;Y|Z)=H(X|Z)+H(Y|Z)-H(X,Y|Z)=H(X|Z)-H(X|Y,Z)=D_{KL}(p(X,Y|Z)\|p(X|Z)p(Y|Z)) \ge 0$, =0 iff $X\perp\!\!\!\perp Y|Z$.
- **MI chain rule**: I(X,Y;Z)=I(X;Z)+I(Y;Z|X). **Data processing inequality**: X−Y−Z Markov ⇒ **I(X;Z) ≤ I(X;Y)** (proof: expand I(X;Y,Z) two ways; I(X;Z|Y)=0). **Invariance**: I(X;Y) invariant under bijections of X,Y (also continuous).
- Venn: H(X)↔μ(A), H(X,Y)↔μ(A∪B), H(X|Y)↔μ(A\B), I↔μ(A∩B). Co-information I(X;Y;Z)=H(XYZ)−H(XY)−H(YZ)−H(XZ)+H(X)+H(Y)+H(Z) can be **negative** (signed measure).
- **Gaussian**: **$\mathbf{I(X;Y)=-\tfrac{1}{2}\ln(1-\rho^2)}$** (ρ correlation). Uncorrelated ≠ independent (MI detects nonlinear dependence).
- **MI & prediction** (mock Q2c): $I(X;Y)=H(Y)-H(Y|X)=H(Y)+E_p[\log p(y|x)] \ge$ **$\mathbf{H(Y)+E_p[\log q_\theta(y|x)]}$** (Gibbs); sample version $H(Y)+(1/n)\Sigma\log q(y_i|x_i)$; equality if $p(y|x)\in Q$ at optimum.
- Helpful: I≥0 ⇒ H(X,Y) ≤ H(X)+H(Y) (equality iff $X\perp\!\!\!\perp Y$). Joint RVs = different partitions of Ω; $p(X=x,Y=y)=p(\omega\in X^{-1}(x)\cap Y^{-1}(y))$ (die: X=even, Y=>3). Comma separates RVs, semicolon separates MI arguments: I(X,Y;Z). Entropy chain-rule proof: p(x,y)=p(x|y)p(y), log splits into H(X|Y)+H(Y).

LECTURE 5 · Noisy channels & codes

Channels & capacity

- **DMC**: input alphabet X, output Y, p(y|x); extended channel $p(y^n|x^n)=\prod_i p(y_i|x_i)$. **BSC** (flip f), uniform input: $H(Y|X)=H_2(f)$, $H(Y)=1 \Rightarrow$ **$I(X;Y)=1-H_2(f)$**.
- **Capacity $C=\max_{p(x)} I(X;Y)$**; maximiser p*(x) = capacity-achieving input distribution. Interpretation: highest rate (bits per channel use) at which info can be sent with arbitrarily small error. Mock Q3: Z-channel p(Y=0|X=0)=1, p(Y=0|X=1)=½; param. p(X=1)=α, compute I(α)=H(Y)−H(Y|X), then dI/dα=0 for α*.
- **Noisy typewriter** (27 letters, each → itself or 2 neighbours w.p. ⅓): H(Y)=log27, H(Y|X)=log3, H(X|Y)=log3, I=log9. Error-free by using **non-confusable** inputs (disjoint output sets, e.g. B,E,H…). Channel AEP: #outputs $2^{nH(Y)}$, confusable per input $2^{nH(Y|X)} \Rightarrow$ non-confusable inputs $2^{nH(Y)}/2^{nH(Y|X)}=\mathbf{2^{nI(X;Y)}}$.

Block codes & the channel coding theorem

- **(n,K) block code**: 2^K codewords of length n; **rate R=K/n** (bits/channel use); p(s)=2^{-K}. Decoder ŝ=Dec(yⁿ) (may output failure). Block error $\mathbf{p_B}=\Sigma p(s)p(\hat{s}\neq s|s)$ (mean); $\mathbf{p_{BM}}=\max_s p(\hat{s}\neq s|s)$ (maximal); $\mathbf{p_b}$ = bit error.
- **Noisy channel coding theorem**: (1) ∀ε>0, R<C, large n ∃ code of rate R with **$p_{BM}<\varepsilon$**. (2) If p_b acceptable, rates up to **$R(p_b)=C/(1-H_2(p_b))$** achievable. (3) For any p_b, rates > $R(p_b)$ **not** achievable.
- **Jointly typical set** $J_\varepsilon^{(n)}$: $|-(1/n)\log p(x^n)-H(X)|<\varepsilon$, same for y^n with H(Y), and (x^n,y^n) with H(X,Y). Properties: (1) $p(x^n,y^n)\approx 2^{-nH(X,Y)}$; (2) $Pr[(x^n,y^n)\in J]\to 1$ (union bound on complements); (3) $|J| \le 2^{n(H(X,Y)+\varepsilon)}$. (4) independent $\tilde{x}^n,\tilde{y}^n \sim p(x^n)p(y^n)$: $Pr[\in J] \le \mathbf{2^{-n(I(X;Y)-3\varepsilon)}}$.
- **Proof of CCT(1), 4 ingredients**: **joint typicality**; **random codes** (average over random codes; if avg < δ some code is < δ – "weighing babies"); **typical-set decoding** (success iff exactly one s with $(x^n(s),y^n)$ jointly typical); **expurgation**. Errors (union bound): $E[p_B] \le Pr[\text{true } x^n(s) \text{ not typical}]$ (→0, property 2) + $(2^{nR}-1)2^{-n(I-3\varepsilon)} \le 2^{-n(I-R-3\varepsilon)}$ (→0 iff **R<I(X;Y)−3ε**; use p* ⇒ R<C−3ε).
- **Expurgation**: take code with mean error p_B, discard worst half of codewords ⇒ remaining have $p_{BM} \le 2p_B$; rate R′=R−1/n → R.

Hamming codes & more

- Error **detection** vs **correction**. R2 detects 1 flip (01→?); R3 corrects 1 flip (101→1). **Parity**=XOR=sum mod 2.
- **Hamming (n,K)**: K data bits, m parity bits, $n=m+K=2^m-1$. **(7,4)**: 3 parity+4 data, 2^4=16 codewords; layout $p_1p_2d_1p_3d_2d_3d_4$ (parity at positions 1,2,4). **$p_1=d_1+d_2+d_4$, $p_2=d_1+d_3+d_4$, $p_3=d_2+d_3+d_4$** (mod 2).
- **$H\in\{0,1\}^{m\times n}$** (parity-check): column j = binary of j. **$G\in\{0,1\}^{K\times n}$** (generator): data columns = identity I_K, parity columns = data rows of H^T. **$\mathbf{t=G^Ts}$** mod 2 (transmitted); **r=t+n** mod 2 (received, noise n); **syndrome z=Hr mod 2**; Ht=0 ⇒ z=H(t+n)=Hn. z=0 ⇒ no error; otherwise z = **binary encoding of the position of the flipped bit** (e.g. error at bit 3 ⇒ $z=(1,1,0)^T \to 1\cdot2^0+1\cdot2^1+0\cdot2^2=3$). Systematic = data first, parity last; (above is non-systematic).
- **Hamming distance** d = # differing bits; Hamming codes have min distance 3. Code with min distance t corrects up to **$\lfloor(t-1)/2\rfloor$** errors. Sphere packing; fully packed ⇒ **perfect** code.
- **Source–channel separation**: optimum error-free rate for sending a process V over p(Y|X) is achieved by a separate source code for V and channel code for Y|X. Summary: **entropy = optimal compression**; **mutual information = optimal transmission**.

LECTURE 6 · Inference & Bayes

Frequentist hypothesis testing

- Descriptive vs inferential statistics. **Frequentist**: a true fixed θ exists; **Bayesian**: beliefs about θ (θ is a random variable).
- **NHST**: H_0 (null, "business as usual") vs H_1; rejection region R: x∈R ⇒ reject H_0. **Simple** hypothesis = point θ=θ_0; **composite** = region θ∈Θ_0; **nested**: H_0 fixes a parameter of H_1 (e.g. a=0 vs a≠0).
- **Type I** (false positive): **α=P(x∈R|H_0)**. **Type II** (false negative): **β=P(x∉R|H_1)**. **Power=1−β**. (Boy who cried wolf: I = wolf claimed falsely.) Can only reject / fail to reject H_0 – never "accept" it.
- **Neyman–Pearson lemma**: for simple H_0:θ=θ_0 vs H_1:θ=θ_1, the **most powerful** test at level α rejects when **$p_{\theta_1}(x)/p_{\theta_0}(x) > k$**, with k s.t. $p_{\theta_0}(x\in R)=\alpha$. (Proof: compare R with another R′ of equal α; R\R′ and R′\R have equal H_0 prob φ; on R\R′ ratio >k, on R′\R ratio <k ⇒ R has more H_1 prob.)
- **Likelihood ratio** $\Lambda=\sup_{\theta\in\Theta_1} p_\theta(x)/\sup_{\theta\in\Theta_0} p_\theta(x)$. Log version $\log\Lambda=\log p_{\theta_1^*}(x)-\log p_{\theta_0^*}(x)$; i.i.d.: $\Sigma_i[\log p_{\theta_1^*}(x_i)-\log p_{\theta_0^*}(x_i)] \to nE[\ldots]$. Optimal for point hypotheses (N–P); Karlin–Rubin: composite with monotone likelihood; otherwise often good but not always optimal. log Λ = "evidence for H_1 over H_0".
- **MI as LR**: H_0: p(x,y)=f(x)g(y) vs H_1: joint. $\sup_{\Theta_1}\log p=-nH(X,Y)$; $\sup_{\Theta_0}\log p=-n(H(X)+H(Y)) \Rightarrow$ **$I(X;Y)=(1/n)E[\log\Lambda]$**. Most info quantities = log-likelihood ratios.

Bayesian inference

- **Bayes**: **$p(\theta|D)=p(D|\theta)p(\theta)/p(D)$**; posterior = likelihood×prior / evidence. Evidence (marginal likelihood) **$p(D)=\int p(D|\theta)p(\theta)d\theta = E_{p(\theta)}[p(D|\theta)]$** (frequentist: sup; "Bayesian playbook: replace sups with sums/integrals").
- **Predictive** $p(D'|D)=\int p(D'|\theta)p(\theta|D)d\theta$ (D−θ−D′ Markov chain). **MAP** θ=argmax p(θ|D)=argmax p(D|θ)p(θ). Gaussian mean (known σ^2, prior $N(\mu_0,\sigma_0^2)$): posterior mean $\mu_1=(\sigma_0^2x+\sigma^2\mu_0)/(\sigma_0^2+\sigma^2)$.
- **Bayesian hypothesis test**: p(H|D)=p(D|H)p(H)/p(D), p(D)=p(D|H_0)p(H_0)+p(D|H_1)p(H_1), where $p(D|H)=\int p(D|\theta,H)p(\theta|H)d\theta$. **Posterior odds $p(H_1|D)/p(H_0|D) = BF \times p(H_1)/p(H_0)$**.
- **Bayes factor BF=p(D|H_1)/p(D|H_0)**: ≫1 ⇒ H_1; ≈1 inconclusive; ≪1 ⇒ H_0 (can support either). Coin (n flips, h heads): H_0:q=½ ⇒ $p(D|H_0)=2^{-n}$; H_1:q~U(0,1) ⇒ $p(D|H_1)=\int q^h(1-q)^{n-h}dq = h!(n-h)!/(n+1)!$. Small n inconclusive; BF→0 if h/n≈½ else ∞.
- **Occam's razor**: evidence ≈ best-fit likelihood × **Occam factor**: $p(D|H)\approx p(D|w_{MP},H)\cdot\sigma_{w|D}/\sigma_w$ (posterior width / prior width). Low when large parameter space but only a tiny part fits ⇒ "a model that can explain everything explains little".
- **MDL** (Shannon's razor): good models give short data descriptions: **$L(D)=L(H_i)+L(w^*|H_i)+L(D|w^*,H_i)$**. Simple model: short L(w*) but long L(D|w*); complex model opposite; best trade-off wins.
- Mock Q4 template: classify test (simple/composite/nested: here H_0,H_1 have free params λ,σ ⇒ composite, not nested); state N–P (LR optimal for simple hypotheses); for i.i.d. data log $\Lambda=\Sigma[\log q_\sigma(x_i)-\log p_\lambda(x_i)]$ then substitute MLEs (θ*=argmax likelihood) and write in terms of x̄ and mean of x².

LECTURE 7 · Estimating info & testing

Estimators (information quantities = density-estimation problem)

- Pipeline: data → density estimation → plug into info formula. Types: 0. plug-in (discrete); I. parametric; II. non-parametric; III. variational.
- **0. Plug-in** (counting): $\hat{p}(x_i)=\#(X=x_i)/\#\text{samples}$; $H=-\Sigma\hat{p}\log\hat{p}$. (Binning continuous data: usually bad idea.)
- **I. Parametric**: fit q_θ, θ*=argmax $q_\theta(x^n)$, then $H(X)=-E_{q^*}[\log q^*(x)]$ analytically (e.g. Gaussian: $\frac{1}{2}\ln|2\pi e\Sigma|$). **Garbage in, garbage out**: a Gaussian fit gives I(X;Y)=0 for dependent but uncorrelated data. Can yield bounds on H/MI.
- **II. Non-parametric**: fit q(x) from data, then evaluate on the data: **$H(X)=-E_p[\log q(x)] \approx -(1/n)\Sigma\log q(x_i)$**. **Kernel** (KDE): bump on each x_i, $K_h(x-x_i)=(2\pi h^2)^{-1/2}\exp(-(x-x_i)^2/2h^2)$, $q(x)=(1/n)\Sigma_i K_h(x-x_i)$; choose h by cross-validation. **k-nearest-neighbour** (Kraskov et al.): adapt width to density; area A containing K neighbours, p locally uniform.
- **III. Variational**: estimate = $\max_\theta$ bound $\hat{H}_\theta(x^n)$ (estimation → optimisation). **(a) Predictor** (Barber–Agakov): **$I(X;Y) \ge H(Y)+E[\log q_\theta(y|x)]$** (any q_θ, e.g. deep net; good for regression/classification). **(b) Contrastive** (density ratio): positive samples $(x_i,y_i)\sim p(x,y)$, negatives $(x_i,y_j)\sim p(x)p(y)$; train classifier f_θ; $I \ge L_{CE}=-E[\log f_\theta(x_i,y_i)/\Sigma_j f_\theta(x_j,y_i)]$; optimal $f=p(x,y)/(p(x)p(y))$.
- **Bias**=$E[\hat{I}]-I$; **Variance**=$E[(\hat{I}-E[\hat{I}])^2]$; both worsen with little data; cheaper methods make more assumptions. Mock Q2(a): Gaussian estimator unsuitable if data non-Gaussian / dependent but uncorrelated.

Hypothesis testing in practice

- **Statistic** T(x); **sufficient** if θ−T−X is a Markov chain. Reject region R={x:T(x)>k}; Λ optimal. Q1: significance testing; Q2: power analysis.
- **p-value** = P(result ≥ as extreme as observed T(x) | H_0). Reject (significant) if p<α (typ. 0.05). **Null distribution** $p_{\theta_0}(T=t)$; critical value: **$\alpha=\int_{t_{crit}}^{\infty} p_{\theta_0}(T=t)dt$**. Report both effect size and p-value (only-p or no-p both bad).
- **Wilks' theorem**: as n→∞, 2 log Λ ~ χ^2 under H_0 (finite n: hard analytically); extension: non-central $\chi^2(\lambda)$ under H_1.
- **Surrogate data**: sample the null empirically by generating fake data where H_0 holds (randomise only the property tested). Independence test $X\perp\!\!\!\perp Y$ with T=Î(X;Y): (1) estimate Î(X;Y); (2) **shuffle Y** → Y^S; (3) compute Î(X;Y^S) over many surrogates; (4) **p = #{Î(X;Y^S) ≥ Î(X;Y)} / #surrogates**. Triplets: statistic – null hypothesis – surrogate method.
- **Power analysis**: power=1−β=P(x∈R|H_1) (true positive rate) ⇒ sample size needed. Simulation: pick n and μ∈H_1; draw n samples from alternative; run NHST at α; repeat M times; **power = fraction rejected**. Increases with effect size and n. **Bias estimation**: fit p(x,y) to data → ground-truth I; draw n samples M times, estimate Î; compare.

LECTURE 8 · Information geometry

Geometry & divergences

- Euclidean: distance $s^2=z^TGz$, $G=I_n$, z=x−y. **Manifold** locally ≈ $\mathbb{R}^n$; $ds^2=dx^Tg(x)dx$ (g = metric; polar coords $g=\text{diag}(1,r^2)$: $ds^2=dr^2+r^2d\varphi^2$). **Statistical manifold**: each point is a PDF.
- **Divergence** D:M×M→[0,∞): (1) D(p,q)≥0; (2) D(p,q)=0 iff p=q; (3) D(p,p+dp) is a positive-definite quadratic form. KL satisfies all (1–2 Gibbs). Others: Bregman, f-divergences.

Projections

- **Projection** = idempotent map P=P∘P. Euclidean orthogonal: $\text{Proj}_M v=\text{argmin}_{w\in M}\|v-w\|^2$.
- **I-projection** $q^*_I=\text{argmin}_{q\in Q}$ **$D_{KL}(q\|p)$**: **zero-forcing** (q=0 where p=0); expectation wrt q – approximate complex with simple. **M-projection** $q^*_M=\text{argmin}_{q\in Q}$ **$D_{KL}(p\|q)$**: **zero-avoiding** (q>0 where p>0); expectation wrt p – usable on real data. Not equal in general (KL asymmetric).
- **ML = M-projection**: $D_{KL}(p\|q)=-H(p)-E_p[\log q] \approx -H(p)-(1/n)\Sigma\log q(x_i) \Rightarrow \text{argmax } E_p[\log q]=\text{argmin } D_{KL}(p\|q)$.
- **MI = M-projection onto factorisable PDFs** F={q:q(x|y)=q(x)}: **$I(X;Y)=\min_{q\in F} D_{KL}(p\|q)$** (min over f,g of $D_{KL}(p\|f(x)g(y))$; by Gibbs optimum f=p(x), g=p(y) ⇒ −H(X,Y)+H(X)+H(Y)). **CMI**: $I(X;Y|Z)=\min_{q\in C} D_{KL}(p\|q)$, C={q:q(x|y,z)=q(x|z)}.
- **Pythagoras**: r = M-projection of p onto Q, q∈Q ⇒ **$D_{KL}(p\|q)=D_{KL}(p\|r)+D_{KL}(r\|q)$** (also for I-projections; needs "dually flat" Q). With F⊂C: I(X;Y,Z)=I(X;Y|Z)+I(X;Z) ⇒ **MI chain rule from Pythagoras**.

Maximum entropy

- **MaxEnt (Jaynes)**: given only $E[f_i(X)]=\alpha_i$, choose p=argmax H(X) s.t. constraints. Lagrange ⇒ **$p(x)=Z^{-1}\exp(-\Sigma_i\lambda_i f_i(x))$** (exponential family; λ_i set so $E[f_i]=\alpha_i$). MaxEnt = **I-projection of the uniform u** on $Q=\{q:E_q[f_i]=\alpha_i\}$ ($D_{KL}(q\|u)=-H(q)+\log u$). More constraints ⇒ lower max entropy: $H(q_2^*)\le H(q_1^*)$.

Metrics: Fisher information

- Taylor: $D_{KL}(\theta\|\theta+d\theta)$ offset=0 (Gibbs), 1st derivative=0 (min) ⇒ **$D_{KL}(\theta\|\theta+d\theta) \approx \frac{1}{2}d\theta^T I(\theta)d\theta$**. **Fisher information $I(\theta)=-E[\partial^2\log p_\theta(X)/\partial\theta^2]$** (vector θ: Hessian $I_{jk}=-E[\partial^2\log p_\theta/\partial\theta_j\partial\theta_k]$) = natural metric; sharpness of likelihood peak; high I ⇒ KL grows fast, more learned about θ.
- Equivalent forms: **$I(\theta)=\text{Var}(\text{Score})$**, score=$\partial\log p_\theta(X)/\partial\theta$; **$I=E[(\partial\log p_\theta/\partial\theta_j)(\partial\log p_\theta/\partial\theta_j)]$** (first derivatives only ⇒ autodiff). Gaussian (μ,σ): **$I=\text{diag}(\sigma^{-2}, 2\sigma^{-2})$**.
- **Bias–variance**: $MSE(\theta)=E[(\theta-\theta)^2]=\text{Bias}^2+\text{Var}$ (cross term vanishes). **Cramér–Rao**: $MSE(\theta) \ge b(\theta)^2+(1+b'(\theta))^2/I(\theta)$, b=Bias ⇒ Fisher info bounds estimation; trade-off has geometric origin.
- **Wrap-up**: same formulae from 3 views – optimal communication (entropy, MI), optimal hypothesis tests (log-LR), projections on statistical manifolds. Tools: info inequalities, LR tests, maxent modelling.